

Complete Data Scientist Roadmap & Syllabus

1. Computer & Programming Basics

- How computers work
- Variables & data types
- Input/output
- Operators
- Conditions (if-else)
- Loops (for, while)
- Functions
- OOP basics
- File handling
- Exception handling
- Python basics

2. Python for Data Science

- Lists, Tuples, Sets, Dictionaries
- Lambda functions
- List comprehension
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Data cleaning
- GroupBy, Merge/Join
- Missing values handling

3. SQL for Data Science

- SELECT, WHERE, ORDER BY
- GROUP BY, HAVING
- CASE WHEN
- Joins (Inner, Left, Right, Full)
- Subqueries
- CTE
- Window functions
- Rank/Dense Rank
- MySQL & PostgreSQL

4. Statistics

- Mean, Median, Mode
- Variance & Standard deviation
- Probability
- Bayes theorem
- Normal, Binomial, Poisson distributions
- Hypothesis testing
- p-value
- t-test, z-test, chi-square test
- ANOVA
- Correlation & Covariance

5. Mathematics for Machine Learning

- Linear Algebra
- Vectors & Matrices
- Eigenvalues
- Calculus
- Derivatives & Gradients
- Gradient Descent

6. Data Analysis & EDA

- Data cleaning
- Handling null values
- Outlier detection
- Feature engineering
- Data transformation
- Visualization

7. Machine Learning

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- SVM
- KNN
- Naive Bayes
- XGBoost

- K-Means clustering
- PCA

8. Model Evaluation

- MAE, MSE, RMSE
- R^2 Score
- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Bias vs Variance
- Overfitting & Underfitting
- Cross Validation

9. Deep Learning

- Neural Networks
- ANN
- CNN
- RNN
- LSTM
- Transformers
- TensorFlow
- Keras
- PyTorch

10. NLP

- Text cleaning
- Tokenization
- Stemming
- Lemmatization
- TF-IDF
- Word embeddings
- BERT
- LLM basics

11. Computer Vision

- Image processing

- CNN
- Object detection
- Image classification
- OpenCV

12. Big Data & Cloud

- Hadoop basics
- Spark
- PySpark
- AWS
- Azure
- GCP

13. Deployment & MLOps

- Flask/FastAPI
- APIs
- Docker
- Git/GitHub
- CI/CD
- Model deployment

14. BI Tools

- Power BI
- Tableau
- Excel dashboards
- KPIs
- Business reporting

15. Projects

- Titanic prediction
- House price prediction
- Customer churn prediction
- Movie recommendation system
- Chatbot
- Stock prediction

16. Interview Preparation

- Python interview questions

- SQL interview questions
- Machine learning concepts
- Case studies
- Problem solving

17. Portfolio Building

- GitHub profile
- LinkedIn profile
- Resume
- Kaggle projects

Approximate Timeline

Stage	Estimated Time
Python + SQL	2 months
Statistics + EDA	1.5 months
Machine Learning	2 months
Projects	2 months
Deep Learning/NLP	2 months
Interview Preparation	Continuous